IHP GUIDE

Integrated Human Practices Guide

Human Practices is how a team finds out what it got wrong before it builds further. This guide turns what we learned from AeroSense's work into steps another team can follow: how to turn feedback from people outside your team into specific, traceable changes to your project. It rests on one project, so treat it as a starting point and adapt it to yours.

1. Write down your assumptions first

Before you talk to anyone, write what you currently believe and what would change your mind. We had two assumptions that were invisible from inside the lab: that our business solved a problem someone would pay to have solved, and that the public misunderstood our topic in the way we had assumed. Both only became visible when we exposed them to outsiders.

2. Choose people by the question they can answer

Group your evidence into loops, and say which loop each piece can answer. We used four: the user, education, the problem, and the technical claim. Our public survey told us about users and what they misunderstand. It could not tell us where the pain sits in the supply chain or what our platform can responsibly claim, and we did not stretch it to. Pick interviewees for the loops you have not yet covered.

3. Start before your first big document

Talk to outsiders before you write the business plan, and before your first conference if you can. Our first outside read came at a conference, and its hardest questions on scale, regulation and cost arrived before we had begun the consultations they led us to add. An earlier conversation with one customer might have saved us two rounds of repositioning.

4. Ask questions that can change your plan

Prepare questions whose answers could overturn what you believe, and keep one standing question for every interview: who actually pays for this? In our case the buyer moved from individual farmers to associations and certifiers only because that question happened to come up. A conversation with an instrument professional overturned a belief we had never examined: we thought we were cheaper than a laboratory instrument, but the instrument's cost was not the barrier.

5. Record what you heard, in their words

Write down what people actually said, not a polished summary, including the blunt parts. Mark any figure an interviewee gave you as theirs, and say whether you have verified it. Keep a single reply from a single company as one data point, and do not read it as proof either way.

6. Turn every piece of feedback into a change

Keep a change log with three columns: what you heard, what it taught you, and what you changed. If an observation did not change anything, say so and say why. Examples from ours:

- A question about regulation at a conference taught us that a diagnostic device faces a much heavier approval path than a screening tool. We moved our first application from medical diagnostics to agriculture.
- A mentor review taught us that a technology that can do many things is not a customer with one expensive problem. We narrowed to a single beachhead.
- A survey taught us that people's gap was a decision habit, not missing facts. We replaced a talk with a decision-based game.

7. Report what is still open

Say which questions you have not answered. Ours included whether customers feel the pain we assumed: one cautious reply, and still open. If outreach does not produce an interview, report it as an attempt that did not succeed. A project that shows its open questions is easier to trust than one with none.

8. Protect and credit the people who helped

Ask for consent before you quote or name anyone. Withhold names and organizations until they confirm, and describe them by role. Thank them, and tell them what you changed because of what they said.

Checklist

☐ We wrote our assumptions down before asking anyone.

☐ We know which loop each piece of evidence can answer.

☐ We talked to outsiders before our first major document.

☐ Each interview had a question that could change our plan.

☐ We recorded what people said and marked unverified figures.

☐ Every piece of feedback is in a change log.

☐ We state what is still open.

☐ Interviewees consented, and we describe them by role.

Our materials

- Our Integrated Human Practices page: https://pinche567.github.io/iGEM-2026-wiki/human-practices.html